Flashcards — CampusX 100 Days of ML | Day 21–30

23 cards | Front: Question | Back: Answer | Sunday revision

Day 21 — **Q1**

What are the 3 cases in bivariate analysis and which plot for each?

ANSWER

Num vs Num → Scatter plot + Correlation heatmap. Num vs Cat → Boxplot (grouped) / Violin plot / Barplot (mean per group). Cat vs Cat → Stacked bar chart / pd.crosstab() + heatmap.

Day 21 — **Q2**

What does Pearson correlation coefficient (r) measure and what are its limitations?

ANSWER

Measures linear relationship strength. Range: -1 to +1. |r| > 0.7 = strong. Limitation: only detects LINEAR relationships. Two variables can have strong non-linear relationship but r ≈ 0.

Day 21 — **Q3**

What is a pair plot and when do you use it?

ANSWER

sns.pairplot(df, hue='target') — plots scatter plots for ALL pairs of numerical columns in one grid. Use it at the start of EDA for a quick overview of all pairwise relationships. hue adds color by class.

Day 22 — **Q4**

What does a Pandas Profiling report contain?

ANSWER

Overview (shape, missing %, duplicates), per-variable stats and distribution, interaction scatter plots, correlation matrices (Pearson, Spearman, Cramér's V), missing value heatmap, and sample rows.

Day 22 — **Q5**

What are 3 important alerts Pandas Profiling shows that you must act on?

ANSWER

High Correlation (>0.9) → drop one feature. High Cardinality → don't use OHE, use target/frequency encoding. Constant column → drop it (zero variance, useless for any model).

Day 23 — **Q6**

Define Feature Engineering and name its 6 types.

ANSWER

Transforming raw data into model-ready features using domain knowledge. Types: Feature Transformation, Feature Encoding, Feature Construction, Feature Splitting, Feature Selection, Feature Scaling.

Day 23 — **Q7**

What is the difference between Feature Engineering and Feature Selection?

ANSWER

Feature Engineering: CREATES or transforms features — adds new information to the dataset. Feature Selection: REMOVES features — keeps only the most useful ones. Both are needed: engineer first, then select.

Day 24 — **Q8**

What is Standardization? Give the formula and output.

ANSWER

StandardScaler: z = (x - mean) / std. Output: mean=0, std=1. No fixed range (≈ -3 to +3 for normal data). Does NOT make data normally distributed — only rescales it.

Day 24 — **Q9**

Which algorithms need scaling and which don't?

ANSWER

Need scaling: KNN, SVM, Logistic Regression, Neural Networks, PCA, K-Means. Don't need: Decision Tree, Random Forest, XGBoost, Naive Bayes. Rule: distance-based and gradient-based algorithms need scaling. Tree-based don't.

Day 25 — **Q10**

What is MinMaxScaler? Formula, output, and when to use.

ANSWER

Formula: x_scaled = (x - min) / (max - min). Output range: [0, 1]. Use for: Neural Networks, when bounded output needed. Sensitive to outliers — one extreme value compresses all others near 0 or 1.

Day 25 **Q11**

What is RobustScaler and why use it?

ANSWER

Formula: (x - Q2) / (Q3 - Q1) — uses median and IQR instead of mean and std. NOT sensitive to outliers. Use when data has significant outliers you can't remove. IQR = Q3 - Q1 is not affected by extreme values, so scaling stays robust.

Day 25 **Q12**

Compare all 4 scalers — when to use each.

ANSWER

StandardScaler: default, symmetric data. MinMaxScaler: Neural Networks, [0,1] needed, no outliers. MaxAbsScaler: sparse data (TF-IDF). RobustScaler: data with outliers. No scaler: tree-based models (RF, XGBoost, DT).

Day 26 **Q13**

What is Ordinal Encoding? When to use and what's the risk?

ANSWER

Assigns integers to categories in a specified order (Low=0, Medium=1, High=2). Use for: ordinal features with natural order (rating, size, education level). Risk: if you use it on nominal features (city, color), it creates a false ordering the model will learn.

Day 26 **Q14**

What is Label Encoding and when should/shouldn't you use it?

ANSWER

Assigns integers alphabetically to categories. USE for: target variable y only (class labels for classification). DO NOT USE for input features X — it implies a false ordering (Mumbai > Delhi is meaningless).

Day 27 **Q15**

What is One Hot Encoding? When to use and what problem does it solve?

ANSWER

Converts a categorical column with N categories into N binary (0/1) columns. Use for: nominal categoricals (no natural order) as input features X. Solves the false-ordering problem of Label Encoding — each category is independent.

Day 27 **Q16**

What is the Dummy Variable Trap and how do you fix it?

ANSWER

With N categories, N binary columns are created. The last column is perfectly predictable from the rest — causing multicollinearity (redundant info). Fix: drop='first' in OneHotEncoder or drop_first=True in pd.get_dummies — creates N-1 columns.

Day 27 **Q17**

What do you do when a categorical column has very high cardinality (500+ unique values)?

ANSWER

Don't use OHE — creates 500+ columns (dimensionality explosion). Use instead: Target Encoding (replace with mean of target per category), Frequency Encoding (replace with count/proportion), or bucket rare categories into 'Other'.

Day 28 **Q18**

What is ColumnTransformer and what problem does it solve?

ANSWER

Applies different sklearn transformers to different column subsets in one step. Problem solved: in real datasets, numerical columns need scaling while categorical columns need encoding — ColumnTransformer handles both cleanly and consistently.

Day 28 **Q19**

What does remainder='passthrough' do in ColumnTransformer?

ANSWER

Columns not mentioned in the transformers list are passed through unchanged (kept as-is). Default is remainder='drop' — unmentioned columns are dropped. Use 'passthrough' when you have columns that don't need any transformation.

Day 29 **Q20**

What is an sklearn Pipeline and why is it important?

ANSWER

Chains multiple preprocessing steps + model into one sklearn object. Ensures same transformations applied to train and test. Prevents data leakage in cross-validation. Deployment-ready — save as one file.

Day 29 | **Q21**

Why is cross-validation wrong without a Pipeline?

ANSWER

Without a Pipeline, if you scale the full dataset before cross-validation, the test fold's statistics affect the scaler — data leakage. Inside a Pipeline, fit() is called fresh for each fold, using only that fold's training data.

Day 30 | **Q22**

What is Log Transform? Formula, when to use, and requirement.

ANSWER

Formula: x_new = log(x) or log1p(x) = log(x+1) for zeros. Use for: right-skewed data (salary, house price, population). Requirement: all values must be positive (> 0). Use log1p if zeros exist. Effect: reduces skewness, compresses large values.

Day 30 | **Q23**

When to use Log vs Sqrt transform?

ANSWER

Sqrt: moderate right skew (0.5–1.0), handles zeros (>= 0 values ok). Log: high right skew (> 1.0), requires positive values (> 0). Log is stronger. Check df['col'].skew() — aim for skewness in [-0.5, +0.5] after transformation.

Rule: Cover the answer, read the question, say the answer out loud, then check. Mark 'known' only if correct without peeking.